

# Aljon A Novelo

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## OBJECTIVE

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- To participate as a **software engineer** in a work environment focused on promoting digital transformation in businesses

## EDUCATIONAL BACKGROUND

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**Bachelor of Science in Computer Science** | *University of the Philippines - Los Baños*  
September 2021 - July 2026

## TECHNICAL SKILLS

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**Languages:** JavaScript, Python, Java, C, Dart, R, Assembly, C++

**Databases:** MongoDB, PostgreSQL, MariaDB, Firebase

**Tools:** GitLab, Google Cloud Platform, RedMine, GitHub, Postman, Figma

## WORK EXPERIENCE

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**Web Developer Intern** | *i-BRIDGE Systems Philippines Inc* | July - August 2025

- Contributed to **Agile sprint management and development workflows** using GitLab, Redmine, and Microsoft Office for collaboration, task tracking, and project documentation
- Designed and implemented **additional user interface components** for an internal tool using the MERN stack
- Developed and integrated **additional database fields** to support the new components
- Conducted **preliminary unit and integration testing** using Mocha and Chai to identify and resolve issues before deployment
- Deployed the new features to **Google Cloud Platform**

## PROJECTS

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**Learning Platform for LaTeX Writers** | *Web Development* | *MERN Stack*

- Conducted **contextual interviews with 20 students** to identify requirements for a LaTeX-focused learning platform
- Developed the platform using the **MERN stack**
- Implemented a **recursive-descent parser** to process LaTeX input
- Achieved a **System Usability Scale score of 8325** and performed **thematic analysis** of interview responses to identify areas for improvement

**Color-Based Image Segmentation** | *Digital Image Processing* | *C++*

- Implemented a **K-means clustering algorithm in C++** to segment images based on dominant color clusters
- Processed **1,000+ sequential JPEG frames**
- Demonstrated the project through a **video presentation** showcasing unsupervised clustering for image segmentation

**Context-Free Grammar Simulator** | *Compiler Design* | *Python*

- Developed a **Python-based command-line simulator** that performs **LL parsing**
- Designed the simulator to support **unambiguous grammars**
- Implemented **file-based input parsing** for grammar definitions

**Computer Architecture Simulation** | *Computer Architecture* | *VHDL*

- Designed and simulated combinational and sequential **circuits** using VHDL, including **test benches** for validating logic behavior

- Evaluated instruction performance and system efficiency through **CPI analysis, speedup computations, and execution time comparisons**
- Conducted benchmark testing to analyze real-world CPU performance, using **emulator tools** and **hardware specification assessments**

#### **Institute Room Reservation System** | *Web Development* | *MERN Stack*

- Implemented dynamic data tables, notifications, and booking workflows for **real-time interaction**
- Integrated **admin and user-specific features**, such as class schedules, announcements, and reports

#### **Networking Game** | *Data Communications and Networking* | *Java*

- Implemented real-time player interactions using **Java sockets** for seamless communication
- Utilized **multithreading** to optimize network performance and ensure responsive gameplay

#### **Programming Language Interpreter** | *Compiler Design* | *Python*

- Built a **parser** that constructs an abstract syntax tree to validate and process LOLCODE programs
- Implemented execution logic for core LOLCODE constructs, including variable assignments, conditionals, loops, and arithmetic operations

### **ORGANIZATIONAL EXPERIENCE**

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#### **UPLB Ugnayan ng Pahinungód**

**Co-Facilitator**, *Maahas Elementary School READ Session* | March 2025

- Prepared learning materials suitable for **Grade 4–6 students**
- Facilitated **peer-to-peer reading sessions**

**Training Participant**, *Training on Mental Health and Psychosocial Support (MHPSS)* | March 2025

- Completed training on **handling MHPSS-related situations in a classroom setting**

**Co-Facilitator**, *HUME 200a Teambuilding Activity* | October 2024

- Facilitated a team-building activity for **undergraduate students from the Department of Human and Family Development Studies** to enhance their interpersonal skills, teamwork, and communication

**Co-Facilitator**, *Batch 42 Volunteer Training Camp* | March 2024

- Co-facilitated training sessions for **59 new volunteers**

**Co-Facilitator**, *CAS NSTP Teambuilding Activity* | October 2023

- Facilitated team-building activities for **an NSTP section** of the College of Arts and Sciences to enhance their group dynamics and equip them with facilitation skills for their literacy projects

**Co-Facilitator**, *Batch 41A and 41B Volunteer Training Camp* | October 2023

- Co-facilitated training sessions for **90 new volunteers**

**Co-Facilitator**, *CEAT NSTP Teambuilding Activity* | October 2023

- Facilitated team-building activities for **5 sections** of NSTP 2 classes of the College of Engineering and Agro-industrial Technology to develop their teamwork and dynamics

**Proctor**, *Los Baños Senior High School CAT Simulation Exam* | May 2023

- Assisted in administering college entrance exam simulations for **Grade 11 students**

#### **UPLB Street Jazz Dance Company**

**Technical Director** | October 2023 – May 2024

- Led a **technical committee** responsible for stage setup, audio, and lighting
- Coordinated with **partner organization** to align production plans
- Managed **budgeting plans** for the committee, including canvassing suppliers, comparing quotations, and negotiating costs

**Logistics Committee Member | October 2022 – October 2023**

- Assisted the group's **logistical needs** during events
- Contributed to finding **venues**, purchasing **event materials**, and creating **set designs**
- Maintained the **properties** of the group

**Kapatirang Pitong Lawa sa UP Los Baños**

**Socio-Cultural Committee Member | August 2024 – July 2026**

- Led the participation of the organization in the annual **UPLB February Protest Fair**
- Managed the **logistics** of booth management
- Handled the **outgoing and incoming finances** of the booth
- **Educated** younger resident members on how to manage the participation of the organization in the protest fair

**Educational Committee Member | August 2023 – June 2024**

- Assisted in the preparation for the **review sessions of Grade 11** students for their upcoming **college entrance exams**

**Executive Member | January – April 2023**

- Led **socio-cultural activities** of the organizations
- Managed **logistical and technical support** for on-ground community activities
- Coordinated with **organization alumni and partner organizations** for activities
- Coordinated with **alumni and event organizers** for internal and external events

**Socio-Cultural Committee Member | August 2022 – January 2023**

- Assisted in **community engagement** activities of the organization

**Membership Committee Member | March – August 2022**

- Assisted in **internal activities** of the organization to promote camaraderie and teamwork

**UPLB College Of Arts and Sciences Freshman Council**

**External Affairs Committee Co-Head | November 2021 – October 2022**

- Handled **incoming and outgoing partnership** requests from university organizations
- Assisted in overseeing the **internal dynamics** of the committee

**External Affairs Member, CASama: An Educational Discussion on Mental Health | April 2022**

- Led the search for **external partnerships** to promote the event
- Monitored the **benefits and responsibilities** of partner organizations

**Project Co-Head, CASinag: A Discussion on Student Leadership | March 2022**

- Managed the **external affairs** of the event
- Monitored the **benefits and responsibilities** of partner organizations
- Coordinated with **guest speakers** to give information and instructions
- Monitored the development of **publicity materials** for the event